

Let $E_i \in \mathcal{M}$, $i \in \mathbb{N}$, and let $\varepsilon > 0$. For every $i \in \mathbb{N}$ there exist $C_i \subset E_i \subset V_i \subset U_n$, C_i compact, V_i open, and since $\mu^* \ll \mu(X)$ is a measure (Carathéodory Thm 4.9),

$$\mu^*(V_i) - \mu^*(C_i) = \mu^*(V_i \setminus C_i) \leq \frac{\varepsilon}{2^{i+1}}.$$

($\downarrow$ decreasing)
 and $\mu^*(\mu(X))$ is a
measure

(N.B. We used the fact that $\mu^*(C_i) < +\infty$). By Thm 4.7

$$\lim_{m \rightarrow \infty} \mu^*\left(\bigcup_{i=1}^{\infty} V_i \setminus \left(\bigcup_{j=1}^m C_j\right)\right) = \mu^*\left(\bigcup_{i=1}^{\infty} V_i \setminus \left(\bigcup_{j=1}^{\infty} C_j\right)\right)$$

(using the fact that $\bigcup_{j=1}^{\infty} V_j \subset U_n$, so $\mu^*\left(\bigcup_{j=1}^{\infty} V_j\right) < +\infty$)

$$\begin{aligned} &\leq \mu^*\left(\bigcup_{i=1}^{\infty} (V_i \setminus C_i)\right) \\ &\leq \sum_{i=1}^{\infty} \mu^*(V_i \setminus C_i) \leq \frac{\varepsilon}{2}. \end{aligned} \quad (*)$$

Find $m \gg 1$ so that

$$\mu^*\left(\bigcup_{i=1}^{\infty} E_i \setminus \underbrace{\left(\bigcup_{j=1}^m C_j\right)}_{\text{compact}}\right) \leq \mu^*\left(\bigcup_{i=1}^{\infty} V_i \setminus \left(\bigcup_{j=1}^m C_j\right)\right) \leq \varepsilon.$$

Hence $\bigcup_{i=1}^{\infty} E_i$ is inner regular, and outer regularity follows from

$$\mu^*\left(\bigcup_{i=1}^{\infty} V_i \setminus \left(\bigcup_{j=1}^{\infty} E_j\right)\right) \leq \mu^*\left(\bigcup_{i=1}^{\infty} V_i \setminus \left(\bigcup_{j=1}^{\infty} C_j\right)\right) \leq \frac{\varepsilon}{2}, \quad (**)$$

because then

$$\begin{aligned} \mu^*\left(\underbrace{\bigcup_{i=1}^{\infty} V_i}_{\text{open}}\right) &= \mu^*\left(\left(\bigcup_{i=1}^{\infty} V_i\right) \setminus \left(\bigcup_{j=1}^{\infty} E_j\right) \cup \bigcup_{j=1}^{\infty} E_j\right) \\ &\leq \mu^*\left(\left(\bigcup_{i=1}^{\infty} V_i\right) \setminus \left(\bigcup_{j=1}^{\infty} E_j\right)\right) + \mu^*\left(\bigcup_{j=1}^{\infty} E_j\right) \end{aligned}$$

$$\leq \mu^*\left(\bigcup_{j=1}^{\infty} E_j\right) + \frac{\varepsilon}{2}. \quad (***)$$

claim If $E \in \mathcal{M}$ then $U_n \setminus E \in \mathcal{M}$,

Fix $\varepsilon > 0$ and find C compact, V open, with
 $C \subset E \subset V \subset U_n$, $\mu^*(V \setminus C) \leq \varepsilon$. (6.1)

Then $U_n \setminus C$ is open, $U_n \setminus E \subset U_n \setminus C$, and
 $\mu^*(U_n \setminus C) = \mu^*(U_n) - \mu^*(C)$ [because $\mu^* \ll \beta(x)$ is a measure
 and $\mu^*(C) < +\infty$]

$$\begin{aligned} &\leq \mu^*(U_n) - \mu^*(V) + \varepsilon \\ &= \mu^*(U_n \setminus V) + \varepsilon \\ &\leq \mu^*(U_n \setminus E) + \varepsilon. \end{aligned}$$

Thus, $U_n \setminus E$ is outer regular.

On the other hand, since U_n is σ -compact there
 exists a compact set $K \subset U_n$ st.

$$\mu^*(U_n) \leq \mu^*(K) + \varepsilon. \quad (6.2)$$

Indeed, write $U_n = \bigcup_{j=1}^{\infty} K_j$, $K_j \subset K_{j+1}$ compact sets,
 and since $\mu^* \ll \beta(x)$ is a measure, by Thm 4.6
 $\mu^*(U_n) = \lim_{j \rightarrow \infty} \mu^*(K_j)$.

Let $K := K_j$ for $j \gg 1$. $j \rightarrow \infty$

Then $K \setminus V$ is compact, $K \setminus V \subset U_n \setminus E$, and by (6.1)

$$\begin{aligned} \mu^*(U_n \setminus E) &\leq \mu^*((U_n \setminus V) \cup (V \setminus E)) \\ &\leq \mu^*((U_n \setminus V) \cup (V \setminus C)) \end{aligned}$$

$$(6.1) \leq \mu^*(U_n \setminus V) + \varepsilon$$

$$= \mu^*((K \setminus V) \cup (U_n \setminus K)) + \varepsilon$$

$$(6.2) \leq \mu^*(K \setminus V) + 2\varepsilon.$$

This shows that $U_n \setminus E$ is inner regular.

Step 2 let $B \in \beta(X)$. We claim that B is outer regular.

Fix $\varepsilon > 0$. By step 1 there exists an open set $V_1 \subset U_1$ such that

$$B \cap U_1 \subset V_1 \quad \text{and} \quad \mu^*(V_1) \leq \mu^*(B \cap U_1) + \frac{\varepsilon}{2}.$$

Inductively, assume that for $k=1, \dots, n$, there exist open sets $V_k \subset U_k$ such that $B \cap U_k \subset V_k$,

$$V_1 \subset V_2 \subset \dots \subset V_n \quad \text{and} \quad \mu^*(V_k) \leq \mu^*(B \cap U_k) + \sum_{i=1}^k \frac{\varepsilon}{2^i}.$$

To construct V_{n+1} , by step 1 find an open set

$$V'_{n+1} \subset U_{n+1} \quad \text{such that}$$

$$(B \cap U_n) \cap U_{n+1} \subset V'_{n+1} \quad \text{and}$$

$$\mu^*(V'_{n+1}) \leq \mu^*((B \cap U_n) \cap U_{n+1}) + \frac{\varepsilon}{2^{n+1}}.$$

Let $V_{n+1} := V_n \cup V'_{n+1}$. Then

$$B \cap U_{n+1} = (B \cap U_n) \cup [(B \cap U_n) \cap U_{n+1}] \subset V_n \cup V'_{n+1} = V_{n+1},$$

$$\begin{aligned} \text{and} \quad \mu^*(V_{n+1}) &\leq \mu^*(V_n) + \mu^*(V'_{n+1}) \\ &\leq \mu^*(B \cap U_n) + \mu^*((B \cap U_n) \cap U_{n+1}) \\ &\quad + \sum_{i=1}^n \frac{\varepsilon}{2^i} + \frac{\varepsilon}{2^{n+1}} \end{aligned}$$

$$= \mu^*(B \cap U_{n+1}) + \sum_{i=1}^{n+1} \frac{\varepsilon}{2^i},$$

where we used the additivity of the measure $\mu^*|_{\beta(X)}$ [Carathéodory Thm 4.9].

Choose that $B \subseteq \bigcup_{n=1}^{\infty} V_n$, an open set, and

$$\begin{aligned} \mu^*\left(\bigcup_{n=1}^{\infty} V_n\right) &= \lim_{n \rightarrow \infty} \mu^*(V_n) \quad (\text{Thm 4.6}) \\ &\leq \lim_{n \rightarrow \infty} \left(\mu^*(B \cap U_n) + \varepsilon \right) \\ &\leq \mu^*(B) + \varepsilon. \end{aligned}$$

Step 3 let $B \in \beta(X)$. we claim that B is inner regular.

let $t < \mu^*(B)$ and choose $n \gg 1$ so that

$$\mu^*(B \cap U_n) > t.$$

By step 1 we can find a compact set $C \subset B \cap U_n$

such that $\mu^*(C) > t$. This shows that $\mu^*(B)$ can be

approximated by the outer measure of compact subsets of B .

Here we used the fact that μ^* is a Borel o.m., hence $\mu^*(B) = \lim_{n \rightarrow \infty} \mu^*(B \cap U_n)$.

Step 4 Assume that μ^* is Borel regular, i.e.,

it is a Borel outer measure ($\beta(X) \subset \mathcal{M}^*$), and

for all $E \subset X \quad \exists B \in \beta(X)$ such that

$$E \subset B, \quad \mu^*(E) = \mu^*(B).$$

We claim that μ^* is a Radon measure, i.e.,

(i) it is a Borel outer measure;

(ii) $\mu^*(K) < \infty$ for every compact set $K \subset X$,
which holds by assumption;

(iii) every open set is inner regular, which
holds by step 3;

(iv) every set $E \subset X$ is outer regular.

Indeed, let $B \in \beta(X)$ be such that $E \subset B$, $\mu^*(E) = \mu^*(B)$

By Step 2 B is outer regular, so given $\varepsilon > 0$ there
exists an open set $V \subset X$, $B \subset V$ and

$$\mu^*(V) \leq \mu^*(B) + \varepsilon.$$

Hence $E \subset V$, and

$$\mu^*(V) \leq \mu^*(B) + \varepsilon = \mu^*(E) + \varepsilon. \quad \blacksquare$$

By Def 6.7, if $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ is a Radon ^{outer} measure, then every set $E \subset X$ is outer regular. In general, we don't have that every set is inner regular, but the following holds:

Proposition 6.11 Let X be a topological space, and let $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ be a Radon measure. Then every σ -finite μ^* -measurable set is inner regular.

Proof Let $E \subset X$ be a μ^* -measurable set, σ -finite.
Step 1 Assume first that $\mu^*(E) < +\infty$.

Since E is outer regular, for $\varepsilon > 0$ there exists an open set $V \supset E$ such that $\mu^*(V) < \mu^*(E) + \varepsilon$.

By the inner regularity of open sets, there exists a compact set $K \subset V$ such that

$$\mu^*(V) < \mu^*(K) + \varepsilon.$$

Since E is μ^* -measurable, we have

$$\underbrace{\mu^*(E \cap V)}_{\mu^*(E)} + \mu^*(V \setminus E) = \mu^*(V) < \mu^*(E) + \varepsilon,$$

and so $\mu^*(V \setminus E) < \varepsilon$. (used $\mu^*(E) < +\infty$)

By outer regularity, there exists an open set $U \supset V \setminus E$ s.t.
 $\mu^*(U) < \varepsilon$. (*)

The compact set $C := K \cup U$ is contained in E , and since U is μ^* -measurable,

$$\begin{aligned} \mu^*(C) &= \mu^*(K \cup U) = \mu^*(K) - \mu^*(K \cap U) \\ &> \mu^*(V) - \varepsilon - \mu^*(K \cap U) \geq \mu^*(V) - \varepsilon - \mu^*(U) \\ &> \mu^*(V) - 2\varepsilon \\ &\stackrel{(*)}{\geq} \mu^*(E) - 2\varepsilon. \end{aligned} \quad \text{Hence } E \text{ is inner regular.}$$

Step 2 If $\mu^*(E) = \infty$, since E is a σ -finite μ^* -measurable set, we can find $E_n \subset E_{n+1}$, $n \in \mathbb{N}$,
and

$$E = \bigcup_{n=1}^{\infty} E_n, \quad \mu^*(E_n) < \infty$$

Note that we don't know that E_n are μ^* -measurable.

By Lemma 6.8, μ^* is Borel regular, so

$\forall n \in \mathbb{N} \quad \exists F_n \in \mathcal{B}(X)$ such that
 $E_n \subset F_n, \quad \mu^*(E_n) = \mu^*(F_n).$

Then $F_n \cap E$ are μ^* -measurable, with

$$\mu^*(F_n \cap E) \leq \mu^*(F_n) = \mu^*(E_n) < \infty,$$

and

$$E = \bigcup_{n=1}^{\infty} (F_n \cap E).$$

We can modify $F_n \cap E$ to make them increasing.

$$(F'_n \cap E, \quad F'_n := \bigcup_{j=1}^n F_j)$$

Hence, by Thm 4.6

$$\infty = \mu^*(E) = \lim_{n \rightarrow \infty} \mu^*(F_n \cap E).$$

Let $M > 0$, choose $n \gg 1$ such that

$$\mu^*(F_n \cap E) > M.$$

Applying step 1 to $F_n \cap E$, we find a compact set

$K \subset F_n \cap E$ such that $\mu^*(K) > M.$

Thus

$$\infty = \mu^*(E) = \sup \{ \mu^*(K) : K \subset E, K \text{ compact} \},$$

hence inner regularity. ■

Remark 6.12

In Proposition 6.10, consider $X = \mathbb{R}^N$ with the Euclidean metric, $\mu^* := (\mathbb{L}^N)^*$.

Then $\mathbb{R}^N$ is a locally compact Hausdorff space, and every open set $A \subseteq \mathbb{R}^N$ is σ -compact. Indeed,

set $A_n := \{x \in \mathbb{R}^N : \text{dist}(x, \mathbb{R}^N \setminus A) \geq 1/n\}$. Then

A_n is closed, so $A_n \cap [-n, n]^N$ is compact and

$$A = \bigcup_{n=1}^{\infty} A_n \cap [-n, n]^N.$$

clearly, $(\mathbb{L}^N)^*(K) < +\infty$ for every K compact, because with $K \subset B(0, R)$, $R > 1$, then

$$(\mathbb{L}^N)^*(K) \leq (\mathbb{L}^N)^*(B(0, R)) \quad (\text{Example 4.10 (i)}) \\ = \alpha^N R^N.$$

By Prop 3.13 $(\mathbb{L}^N)^*$ is a Borel outer measure, since $\mathcal{B}(\mathbb{R}^N) \subset \mathcal{L}(\mathbb{R}^N)$ (σ -algebra of Lebesgue measurable sets).

By Prop 6.10 we have that:

Every Borel set is inner regular and outer regular. (6.3)

(See Cor 6.15)

If we knew that $(\mathbb{L}^N)^*$ is Borel regular, i.e., for all $E \subseteq \mathbb{R}^N$ $\exists B \in \mathcal{B}(\mathbb{R}^N)$ st. $E \subset B$,

$$(\mathbb{L}^N)^*(E) = (\mathbb{L}^N)^*(B) \quad (6.4)$$

then Prop 6.10 $\Rightarrow (\mathbb{L}^N)^*$ is a Radon outer measure. (6.5)

Then, by Prop 6.11 and because of $E \in \mathcal{L}(\mathbb{R}^N)$ then

E is σ -finite $\subset E = \bigcup_{n=1}^{\infty} E \cap [-n, n]^N$, $\mathbb{L}^N(E \cap [-n, n]^N) < +\infty$,

and we would have that

every Lebesgue measurable set is inner regular. (6.6)

We will see that (6.3) holds for every Lebesgue measurable set E , and every set is outer regular, i.e., (6.5).

Thm 6.13 If $E \subseteq \mathbb{R}^N$ then

$$(\mathcal{L}^N)^*(E) = \inf \{ \mathcal{L}^N(U) : E \subset U, U \text{ open} \} \text{ (outer-regularity),}$$

and if E is Lebesgue measurable, then

$$\mathcal{L}^N(E) = \sup \{ \mathcal{L}^N(K) : K \subset E, K \text{ compact} \} \text{ (inner-regularity) (i.e., (6.6)).}$$

In particular, $(\mathcal{L}^N)^*$ is a Radon measure.

Proof step 1 let $E \subseteq \mathbb{R}^N$. We prove that

$$(\mathcal{L}^N)^*(E) = \inf \{ \mathcal{L}^N(U) : E \subset U, U \text{ open} \}.$$

If $(\mathcal{L}^N)^*(E) = +\infty$ then this is clearly true, take $U := \mathbb{R}^N$.

Suppose that $(\mathcal{L}^N)^*(E) < +\infty$. By monotonicity,

$$(\mathcal{L}^N)^*(E) \leq \inf \{ \mathcal{L}^N(U) : E \subset U, U \text{ open} \},$$

we show the reverse inequality,

$$(\mathcal{L}^N)^*(E) \geq \inf \{ \mathcal{L}^N(U) : E \subset U, U \text{ open} \}. \quad (6.7)$$

let $\varepsilon > 0$, and find a covering of E , $\{R_i : i \in \mathbb{N}\}$, by rectangles such that

$$\sum_{i=1}^{\infty} \mathcal{L}^N(R_i) \leq (\mathcal{L}^N)^*(E) + \frac{\varepsilon}{2}.$$

let S_i be a rectangle whose interior S_i° contains R_i , and

$$\mathcal{L}^N(S_i) \leq \mathcal{L}^N(R_i) + \frac{\varepsilon}{2^{i+1}}.$$

Then the collection of open rectangles $\{S_i^\circ : i \in \mathbb{N}\}$ covers E

and with $U := \bigcup_{i=1}^{\infty} S_i^\circ$, we have

$$\mathcal{L}^N(U) \leq \sum_{i=1}^{\infty} \mathcal{L}^N(S_i) \leq \sum_{i=1}^{\infty} \mathcal{L}^N(R_i) + \frac{\varepsilon}{2} \leq (\mathcal{L}^N)^*(E) + \varepsilon,$$

and this proves (6.7).